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Credit Supply and Monetary Policy: Identifying the Bank Balance-Sheet Channel with Loan Applications

Contents

1	Big picture	3
2	Data and measurement (key objects)	3
2.1	Observed loan-application panel	3
2.2	Observed bank and firm data	4
2.3	Key measured objects	4
2.4	Unobservables and timing	5
2.5	Empirical applications	5
3	Models and empirical specifications (all equations + interpretation)	5
3.1	Hypotheses	5
3.2	Baseline loan-granting equation	6
3.3	Why the standard macro or bank-level approach fails	6
3.4	Fixed-effects strategy and increasingly demanding designs	6
3.5	Why linear probability models are preferred	7
3.6	Credit-substitution regressions	7
3.7	Bank relationships as a mitigation channel	8
3.8	Economic interpretation of the interaction coefficients	8
3.9	Robustness and extensions	8
4	Identification and instruments (what makes the estimates credible)	8
4.1	What fails in a naive interpretation	8
4.2	Why Spain helps separate monetary from GDP	9
4.3	Why loan applications help identify supply	9

4.4	Why firm-month and loan fixed effects are decisive	9
4.5	Why the substitution tests matter	9
4.6	Relation to the literature	10
5	Tables and figures	10
5.1	Table 1: Variable definitions and summary statistics	10
5.2	Table 2: Baseline regressions on loan granting	11
5.3	Table 3: Firm-month and loan fixed-effects evidence	11
5.4	Table 4: Summary statistics for the substitution tests	12
5.5	Table 5: Future loan granting after a rejection	13
5.6	Table 6: Future loan granting after an earlier application and the role of relationships	13
5.7	Robustness and extensions	14
6	Short summary	15
7	Conclusion	15
7.1	Core conclusion (what the paper establishes)	15
7.2	Economic intuition	15
7.3	Takeaway	15

1 Big picture

- **Question.** Do tighter monetary policy and adverse economic conditions reduce bank loan supply, and is this contraction stronger for banks with weaker balance sheets?
- **Setting.** The paper studies the Spanish credit market using individual *loan applications* rather than only granted loans. This allows the authors to observe loan demand directly and to ask which bank grants credit to a given borrower under given macro conditions.
- **Core challenge.** Identifying a bank lending channel requires separating *loan supply* from *loan demand*. It also requires separating the effects of monetary conditions from the effects of economic activity.
- **Main contribution.** The paper solves these problems with a uniquely rich administrative dataset from the Credit Register of Spain and with very demanding fixed-effects designs, especially **firm-month fixed effects** and **loan fixed effects**.
- **Main result.** Higher short-term interest rates and lower GDP growth reduce the probability that a loan application is granted. The contraction is stronger for banks with low capital and, especially for monetary tightening, low liquidity.
- **Economic magnitude.** A 100-basis-point increase in the short-term interest rate reduces loan granting by weak banks about **11.4 percent more** than by strong banks. A 100-basis-point decrease in GDP growth reduces loan granting by weak banks about **5 percent more** than by strong banks.
- **Credit-crunch result.** Within the set of applications made by the *same firm in the same month* to different banks, banks with weaker balance sheets grant fewer loans when monetary or economic conditions tighten. This is the paper’s cleanest evidence that a deterioration in bank balance sheets can generate a credit crunch.
- **Substitution result.** Firms cannot simply offset a rejection by turning to other banks, especially in tight monetary and economic conditions. Multiple banking relationships help, but they do not eliminate the effect.
- **Why this paper matters.** It is one of the clearest micro-level demonstrations of the bank balance-sheet channel of monetary policy. The paper shows that not just the policy rate itself, but also the condition of bank balance sheets, determines the supply of credit.

2 Data and measurement (key objects)

2.1 Observed loan-application panel

- **Main data source.** The paper uses the **Credit Register of Spain (CIR)**, managed by the Banco de España.
- **Coverage.** The CIR includes virtually all business loans above 6,000 euros granted by banks operating in Spain. More than 350 banks and more than 130,000 firms are active in the register at any moment in time.

- **Loan type.** The analysis focuses on commercial and industrial loans to nonfinancial corporations. These loans account for about 82% of granted loans in the register.
- **Applications.** The key innovation is that the CIR also records *information requests* filed by banks on potential borrowers. A bank automatically receives information on its current borrowers, so it requests information only when a firm is *not currently borrowing from that bank*. These requests therefore correspond to the **extensive margin of new lending**.
- **Period.** Loan-application requests are observed from 2002:02 to 2008:12. The benchmark sample ends before the extraordinary ECB interventions that began late in 2008, though the paper extends the sample to 2010:06 in robustness exercises.
- **Matching applications to grants.** A loan application is considered granted if a loan from that bank appears in the register in month t to $t + 3$. The paper also checks one- and two-month matching windows and finds similar results.
- **Sample sizes.** The broadest sample contains 2,335,321 loan applications. When applications are matched to firm-level balance-sheet data, the starting estimation sample contains 816,852 applications.

2.2 Observed bank and firm data

- **Bank data.** Monthly balance-sheet information for banks is available from the Banco de España supervisory files beginning in 1984.
- **Firm data.** Firm balance-sheet information comes from the Spanish Mercantile Register at annual frequency starting in 1992.
- **Institutions.** The bank sample includes commercial banks, savings banks, and credit cooperatives, which together account for more than 95% of the Spanish financial system.
- **Borrowers.** The firm sample includes nonfinancial publicly limited and limited liability companies, which account for the vast majority of firms in Spain.

2.3 Key measured objects

- **Main dependent variable.** LOAN APPLICATION IS GRANTED equals one if application (i, b, t) is approved and a loan is granted by bank b to firm i between month t and $t + 3$, and zero otherwise. Its mean is 0.42.
- **Macroeconomic conditions.**
 - ΔIR_t : annual change in the Spanish three-month interbank rate;
 - ΔGDP_t : annual real GDP growth;
 - ΔCPI_t : annual inflation.
- **Key bank balance-sheet variables.**
 - BANK CAPITAL RATIO: core capital over total assets;
 - BANK LIQUIDITY RATIO: liquid assets over total assets.

- **Other bank controls.** Log total assets, ROA, doubtful loan ratio, and the Herfindahl index of the bank's credit portfolio across industries.
- **Firm controls.** Firm capital ratio, liquidity ratio, total assets, age, ROA, current doubtful-loan indicator, lagged doubtful-loan indicator, the duration of the relationship with the bank, and the number of bank relationships.
- **Industry and province controls.** Industry doubtful-loan ratios and the number of banks in the province.
- **Substitution outcomes.**
 - FUTURE LOAN APPLICATION IS GRANTED FOLLOWING REJECTION: whether a firm obtains at least one granted loan in the next three months after a rejected application;
 - FUTURE LOAN APPLICATION IS GRANTED AFTER EARLIER APPLICATION IS MADE: whether a firm gets any bank loan in the next three months after making an application.

2.4 Unobservables and timing

- **Unobserved firm heterogeneity.** Loan demand and borrower quality vary over time because of changes in risk, investment opportunities, internal funds, and outside financing options.
- **Timing.** Bank balance-sheet variables are lagged one month, since capital and liquidity may themselves respond to macro conditions.
- **Identification timing.** The most demanding designs compare the success of applications made by the same firm in the same month to banks with different balance-sheet strength.
- **Monetary-policy environment.** Spain is treated as facing relatively exogenous monetary policy because, during the sample period, its business cycle was imperfectly synchronized with the core euro-area economies setting common policy.
- **Current-borrower limitation.** Applications to a firm's current lenders are not directly observed in the application dataset. The paper therefore adds exercises on firms with no existing bank debt and on post-rejection substitution to show that this does not overturn the main message.

2.5 Empirical applications

- **Main application.** The paper studies the extensive margin of new lending using loan applications.
- **Additional application.** In unreported robustness, it studies the intensive margin using all granted loans in Spain from 1988:II to 2008:IV and finds similar patterns.

3 Models and empirical specifications (all equations + interpretation)

3.1 Hypotheses

The theory yields two central hypotheses:

- **H1.** Higher short-term interest rates or lower GDP growth contract credit availability.
- **H2.** The negative effect of higher interest rates or lower GDP growth on credit availability is stronger for banks with lower capital or liquidity.

3.2 Baseline loan-granting equation

The paper’s baseline empirical specification is

$$\begin{aligned} Grant_{ibt} = & \beta_I \Delta IR_t + \beta_G \Delta GDP_t + \beta_{IC}(\Delta IR_t \times Cap_{b,t-1}) + \beta_{IL}(\Delta IR_t \times Liq_{b,t-1}) \\ & + \beta_{GC}(\Delta GDP_t \times Cap_{b,t-1}) + \beta_{GL}(\Delta GDP_t \times Liq_{b,t-1}) + \gamma' Macro_t + \delta' Bank_{b,t-1} \\ & + \phi' Firm_{i,t-1} + \psi' Other_{t-1} + \varepsilon_{ibt}. \end{aligned} \quad (1)$$

- **Outcome.** $Grant_{ibt}$ is the loan-granting indicator for firm i , bank b , and month t .
- **Macro effects.** $\beta_I < 0$ and $\beta_G > 0$ correspond to Hypothesis 1.
- **Bank balance-sheet channel.** $\beta_{IC} > 0$, $\beta_{IL} > 0$, $\beta_{GC} < 0$, and $\beta_{GL} < 0$ correspond to Hypothesis 2.
- **Interpretation.** A positive β_{IC} , for example, means that high bank capital dampens the negative effect of an interest-rate increase on loan granting.

3.3 Why the standard macro or bank-level approach fails

- Regressions on granted loans alone cannot distinguish a decline in loan *supply* from a decline in loan *demand*.
- Bank-level regressions may also be contaminated because firms facing tighter conditions can change which banks they approach.
- Short-term interest rates may be tied to the business cycle, making it hard to separate monetary from real conditions in many countries.
- The paper addresses all three issues by observing applications, exploiting Spain’s monetary-policy setting, and saturating the regressions with high-dimensional fixed effects.

3.4 Fixed-effects strategy and increasingly demanding designs

The paper estimates progressively more demanding versions of equation (1):

$$Grant_{ibt} = \text{RHS of (1)} + \alpha_i + \varepsilon_{ibt}, \quad (2a)$$

$$Grant_{ibt} = \text{RHS of (1)} + \alpha_i + \lambda_t + \varepsilon_{ibt}, \quad (2b)$$

$$Grant_{ibt} = \text{RHS of (1)} + \alpha_{it} + \varepsilon_{ibt}, \quad (2c)$$

$$Grant_{ib\ell t} = \text{RHS of (1)} + \alpha_\ell + \varepsilon_{ib\ell t}. \quad (2d)$$

- **Firm fixed effects** (α_i) absorb time-invariant borrower heterogeneity.

- **Month fixed effects** (λ_t) absorb aggregate macro conditions; once these are included, the identification comes only from the interaction terms.
- **Firm-month fixed effects** (α_{it}) absorb all observed and unobserved borrower characteristics that vary by month. This is the paper’s benchmark specification.
- **Loan fixed effects** (α_ℓ) compare different banks evaluating the *same loan* within the same month, when at most one bank grants that loan.
- **Economic meaning.** The firm-month and loan fixed-effect specifications implement the Bernanke–Lown notion of a credit crunch by holding borrower quality fixed while allowing bank balance-sheet strength to differ.

3.5 Why linear probability models are preferred

- Fixed effects in nonlinear models such as logit mechanically drop firms that have no within-group variation in outcomes.
- To avoid that selection problem, the paper uses linear probability models in the main analysis.
- An additional advantage is that coefficients on interaction terms are directly interpretable and their standard errors do not require nonlinear interaction corrections.
- Logit specifications are estimated in robustness and give very similar answers.

3.6 Credit-substitution regressions

To study whether rejected borrowers can undo a supply contraction by applying elsewhere, the paper estimates regressions that mirror equation (1) but replace the dependent variable with future lending outcomes. Schematically,

$$FutureGrantAfterReject_{ibt} = \Theta(\Delta IR_t, \Delta GDP_t, Cap_{b,t-1}, Liq_{b,t-1}, \overline{Bank}_{i,t-1}, Controls) + u_{ibt}, \quad (3a)$$

$$AnyFutureGrant_{ibt} = \Lambda(\Delta IR_t, \Delta GDP_t, Cap_{b,t-1}, Liq_{b,t-1}, Controls) + v_{ibt}. \quad (3b)$$

- **First outcome.** Whether the firm receives any granted loan in the next three months after a rejection.
- **Second outcome.** Whether the firm receives any granted loan in the next three months after making an application.
- **Additional controls.** These specifications include the balance-sheet characteristics of the assessing bank and of the average bank approached by the firm.
- **Interpretation.** If firms could frictionlessly substitute across lenders, the main bank-lending-channel estimates might overstate the effective contraction of credit. These regressions test exactly that possibility.

3.7 Bank relationships as a mitigation channel

The paper then interacts monetary and economic conditions, bank balance-sheet strength, and the number of bank relationships:

$$\begin{aligned} AnyFutureGrant_{ibt} = & \Lambda(\cdot) + \rho_1(\Delta IR_t \times Rel_{it-1}) + \rho_2(\Delta IR_t \times Cap_{b,t-1} \times Rel_{it-1}) \\ & + \rho_3(\Delta IR_t \times Liq_{b,t-1} \times Rel_{it-1}) + \rho_4(\Delta GDP_t \times Rel_{it-1}) + \dots \end{aligned} \quad (4)$$

- **Interpretation.** Existing banking relationships may soften the impact of weak bank balance sheets by giving the borrower better outside options or easier access to alternative lenders.
- **Main result.** The bank lending channel is weaker for firms with more bank relationships, but it does not disappear.

3.8 Economic interpretation of the interaction coefficients

The differential effect of a monetary tightening on weak versus strong banks can be summarized as

$$\Delta Grant_{IR}^{weak-strong} = \Delta IR \left[\beta_{IC}(Cap_L - Cap_H) + \beta_{IL}(Liq_L - Liq_H) \right]. \quad (5)$$

Similarly, the differential effect of lower GDP growth is

$$\Delta Grant_{GDP}^{weak-strong} = \Delta GDP \left[\beta_{GC}(Cap_L - Cap_H) + \beta_{GL}(Liq_L - Liq_H) \right]. \quad (6)$$

- **Interpretation.** The paper evaluates these formulas at the tenth and ninetieth percentiles of bank capital and liquidity to produce economically meaningful semielasticities.
- **Main message.** The bank lending channel is not merely statistically significant; the differential effects are quantitatively large.

3.9 Robustness and extensions

- The results survive adding month fixed effects, firm-month fixed effects, loan fixed effects, and in robustness bank fixed effects.
- They are robust to extending the sample to 2010:06, to using logit models, to left-censoring bank capital ratios, to dropping GDP interactions, and to adding interactions with other bank characteristics.
- The paper also reports that intensive-margin evidence using all granted business loans in Spain points in the same direction.

4 Identification and instruments (what makes the estimates credible)

4.1 What fails in a naive interpretation

- **Loan demand is endogenous.** Tighter monetary policy and weaker economic conditions may reduce both the willingness of banks to lend and the desire of firms to borrow.

- **Borrower composition changes over time.** Different types of firms may approach different banks over the cycle.
- **Monetary policy and GDP may move together.** In many settings, the short rate is tightly linked to domestic business-cycle conditions.

4.2 Why Spain helps separate monetary from GDP

- Spain is a peripheral euro-area economy that does not perfectly determine common monetary policy.
- During the sample period, its business cycle was only weakly synchronized with the core countries of the euro area.
- This creates useful variation to distinguish changes in short-term rates from changes in Spanish GDP growth.

4.3 Why loan applications help identify supply

- Observing a loan application means loan demand is directly observed at the bank-firm level.
- Conditional on an application, the bank decides whether to grant or deny credit.
- Because requests are filed only for noncurrent borrowers, the paper isolates the extensive margin of new credit supply rather than automatic renewals or evergreen loans.

4.4 Why firm-month and loan fixed effects are decisive

- Firm-month fixed effects compare applications made by the same firm in the same month to different banks.
- Loan fixed effects compare different banks evaluating the same loan in the same month.
- These designs absorb all borrower-side factors that vary within the month and leave only cross-bank differences in balance-sheet strength to explain differential loan granting.
- This is what lets the paper interpret the estimates as shifts in *loan supply* rather than changes in borrower demand or quality.

4.5 Why the substitution tests matter

- Even if some banks reduce supply, aggregate firm credit might be unaffected if borrowers can easily turn to other banks.
- By studying post-rejection and post-application outcomes, the paper shows that this substitution is limited.
- Therefore, the identified bank-level supply contraction matters for actual firm access to credit.

Variable	Units	Definition	Mean	SD	Min	Median	Max
<i>Dependent variable</i>							
<i>LOAN APPLICATION IS GRANTED_{it}</i>	0/1	= 1 if the loan application by a firm is approved and the loan is granted by a bank, = 0 otherwise	0.42	0.49	0	0	1
<i>Independent variables</i>							
<i>Macroeconomic conditions (f)</i>							
$\Delta\%_t$	%	Annual change of Spanish 3-month interbank interest rates	0.19	0.83	-1.56	0.19	1.41
ΔGDP_t	%	Annual change of Spanish gross domestic product in real terms	3.13	0.93	-0.85	3.35	3.98
ΔCPI_t	%	Annual change of Spanish Consumer Price Index	3.33	0.77	1.43	3.40	5.27
<i>Bank characteristics (b)</i>							
<i>BANK CAPITAL RATIO_{it-1}</i>	%	The ratio of bank equity over total assets of the bank	5.37	2.07	0.00	4.82	63.15
<i>BANK LIQUIDITY RATIO_{it-1}</i>	%	The ratio of liquid assets (cash and balance with central banks, and loans and advances to governments and credit institutions) held by the bank over the total assets of the bank	17.02	8.03	0.04	15.83	92.07
$\ln(TOTAL ASSETS_{it-1})$	—	The log of the total assets of the bank	17.39	1.47	9.57	17.61	19.90
<i>TOTAL ASSETS_{it-1}</i>	000,000,000 EUR	The total assets of the bank	78.00	87.60	0.01	44.20	437.00
<i>ROA_{it-1}</i>	%	The total net income over assets of the bank	0.94	0.55	-8.93	0.91	11.92
<i>DOUBTFUL LOANS RATIO_{it-1}</i>	%	The doubtful loan ratio of the bank	0.83	0.85	0.00	0.56	31.24
<i>HERFINDAHL BY INDUSTRY_{it-1}</i>	%	The Herfindahl-Hirschman index of the bank's credit portfolio by industry	26.35	8.86	12.77	23.20	87.94
<i>Firm characteristics (f)</i>							
<i>FIRM CAPITAL RATIO_{it-1}</i>	%	The ratio of own funds over total assets of the firm	24.52	20.73	0.00	18.67	100
<i>FIRM LIQUIDITY RATIO_{it-1}</i>	%	The ratio of current assets over total assets of the firm	41.14	26.91	0.00	38.26	100
<i>TOTAL ASSETS_{it-1}</i>	000,000 EUR	The total assets of the firm	6.98	75.95	0.00	1.32	27,200.00
$\ln(TOTAL ASSETS_{it-1})$	—	The log of the total assets of the firm	7.26	1.62	0.86	7.19	17.12
<i>AGE_{it-1}</i>	years	The age of the firm	10.30	9.25	0	8	132
$\ln(1 + AGE_{it-1})$	—	The log of one plus the age of the firm	2.10	0.86	0.00	2.20	4.89
<i>ROA_{it-1}</i>	%	The return on assets of the firm	6.66	9.74	-36.07	4.89	63.16
<i>I(DOUBTFUL LOANS AT THE TIME OF THE REQUEST_{it-1})</i>	0/1	= 1 if the firm had doubtful loans the month before the loan was requested, = 0 otherwise	0.00	0.09	0	0	1
<i>I(DOUBTFUL LOANS BEFORE THE TIME OF THE REQUEST_{it-1})</i>	0/1	= 1 if the firm had doubtful loans before the previous month to the loan request, = 0 otherwise	0.09	0.29	0	0	1
<i>NUMBER OF MONTHS WITH THE BANK_{it-1}</i>	months	The duration of the relationship between firm and bank	7.84	23.48	0	0	278
$\ln(1 + \text{NUMBER OF MONTHS WITH THE BANK}_{it-1})$	—	The log of one plus the duration of the relationship between firm and bank	0.63	1.36	0	0	5.63
<i>NUMBER OF BANK RELATIONSHIPS_{it-1}</i>	—	The number of banks the firm is currently borrowing from	3.90	3.66	0	3	102
$\ln(1 + \text{NUMBER OF BANK RELATIONSHIPS}_{it-1})$	—	The log of the number of bank relationships of the firm	1.35	0.65	0	1.39	4.63
<i>Industry characteristic (i)</i>							
<i>INDUSTRY DOUBTFUL LOANS RATIO_{it-1}</i>	%	The doubtful loan ratio of the industry in which the firm operates	0.91	0.60	0.06	0.73	4.91
<i>Province characteristic (p)</i>							
<i>NUMBER OF BANKS_{it-1}</i>	—	The number of banks in the province where the firm is located	116.52	32.52	11	111	179
$\ln(\text{NUMBER OF BANKS}_{it-1})$	—	The log of the number of banks in the province where the firm is located	4.72	0.29	2.40	4.71	5.19

Notes: The number of observations equals 816,852, t : 1 to 83, b : 1 to 184, i : 1 to 267,618, x : 1 to 10, and p : 1 to 52. EUR = 2008 euros.

4.6 Relation to the literature

- Relative to the macro and bank-level literature on the lending channel, the paper contributes by directly observing loan applications.
- Relative to work such as Kashyap and Stein, it uses cleaner measures of bank strength and much tighter borrower-side controls.
- Relative to studies identifying bank shocks with bank-firm data, it is distinctive in its use of **firm-month fixed effects with loan applications**, which is the heart of the paper's identification argument.

5 Tables and figures

5.1 Table 1: Variable definitions and summary statistics

Read:

- The average probability that a loan application is granted is **42 percent**, so denial is common.
- Banks in the sample have an average capital ratio of about **5.4 percent** and an average liquidity ratio of about **17 percent**, with substantial cross-sectional variation.
- Firms are heterogeneous in balance-sheet strength and bank relationships: the average firm borrows from about **3.9 banks**.
- The table matters because the interaction-based identification strategy relies on real dispersion in bank capital and liquidity.

5.2 Table 2: Baseline regressions on loan granting

TABLE 2—REGRESSION RESULTS, LOAN GRANTING AND MONETARY CONDITIONS
(Dependent variable: LOAN APPLICATION IS GRANTED_{it})

Model	(1)	(2)	(3)
Macroeconomic controls (<i>r</i>)			
ΔIR_t	-1.394*** (0.254)	-5.960*** (0.625)	
$\Delta IR_t \times BANK\ CAPITAL\ RATIO_{it-1}$		33.384*** (6.973)	29.991*** (6.979)
$\Delta IR_t \times BANK\ LIQUIDITY\ RATIO_{it-1}$		15.396*** (1.951)	13.343*** (2.044)
ΔGDP_t	4.734*** (0.256)	6.769*** (0.552)	
$\Delta GDP_t \times BANK\ CAPITAL\ RATIO_{it-1}$		-28.580*** (6.725)	-28.728*** (6.539)
$\Delta GDP_t \times BANK\ LIQUIDITY\ RATIO_{it-1}$		-3.340 (2.433)	-2.916 (2.525)
ΔCPI_t	-0.081 (0.183)	-0.027 (0.183)	
Bank characteristics (<i>b</i>)			
$BANK\ CAPITAL\ RATIO_{it-1}$	-0.671*** (0.058)	0.225 (0.218)	0.187 (0.214)
$BANK\ LIQUIDITY\ RATIO_{it-1}$	-0.069*** (0.017)	0.032 (0.080)	0.011 (0.084)
$\ln(TOTAL\ ASSETS_{it-1})$	0.001 (0.001)	0.001 (0.001)	0.001 (0.001)
ROA_{it-1}	0.447 (0.275)	0.473* (0.267)	0.758*** (0.259)
$DOUBTFUL\ LOANS\ RATIO_{it-1}$	0.307** (0.154)	0.355** (0.151)	0.237 (0.146)
$HERFINDAHL\ BY\ INDUSTRY_{it-1}$	0.031* (0.016)	0.057*** (0.016)	0.075*** (0.016)
Firm characteristics (<i>i</i>)			
$FIRM\ CAPITAL\ RATIO_{it-1}$	0.015* (0.009)	0.015* (0.009)	0.019** (0.009)
$FIRM\ LIQUIDITY\ RATIO_{it-1}$	-0.003 (0.005)	-0.003 (0.005)	-0.003 (0.005)
$\ln(TOTAL\ ASSETS_{it-1})$	0.000 (0.002)	0.000 (0.002)	0.001 (0.002)
$\ln(1 + AGE_{it-1})$	0.020*** (0.005)	0.018*** (0.005)	0.026*** (0.004)
ROA_{it-1}	0.082*** (0.010)	0.083*** (0.010)	0.083*** (0.010)
$I(DOUBTFUL\ LOANS\ AT\ THE\ TIME\ OF\ THE\ REQUEST_{it-1})$	-0.092*** (0.009)	-0.092*** (0.009)	-0.090*** (0.009)
$I(DOUBTFUL\ LOANS\ BEFORE\ THE\ TIME\ OF\ THE\ REQUEST_{it-1})$	-0.037*** (0.007)	-0.037*** (0.007)	-0.033*** (0.007)
$\ln(1 + NUMBER\ OF\ MONTHS\ WITH\ THE\ BANK_{it-1})$	0.006*** (0.001)	0.007*** (0.001)	0.007*** (0.001)
$\ln(1 + NUMBER\ OF\ BANK\ RELATIONSHIPS_{it-1})$	-0.162*** (0.003)	-0.162*** (0.003)	-0.161*** (0.003)
Industry characteristic (<i>s</i>)			
$INDUSTRY\ DOUBTFUL\ LOANS\ RATIO_{it-1}$	-0.597*** (0.193)	-0.712*** (0.192)	-1.109*** (0.180)

TABLE 2—REGRESSION RESULTS, LOAN GRANTING AND MONETARY CONDITIONS
(Dependent variable: LOAN APPLICATION IS GRANTED_{it})(Continued)

Model	(1)	(2)	(3)
Province characteristic (<i>p</i>)			
$\ln(NUMBER\ OF\ BANKS_{pt-1})$	0.109*** (0.014)	0.110*** (0.014)	0.107*** (0.014)
Firm fixed effects	Yes	Yes	Yes
Month fixed effects	No	No	Yes
Observations	816,852	816,852	816,852
Number of bank-month clusters	9,910	9,910	9,910
Sample period	2002:02– 2008:12	2002:02– 2008:12	2002:02– 2008:12

Notes: The table reports the estimated coefficients and robust standard errors (in parentheses) clustered at the bank-month level from linear probability models estimated using least squares. Fixed effects are included ("yes") or not included ("no"). The set of month fixed effects includes a fixed effect for every (but one) year:month during the sample period. The variable definitions and summary statistics are in Table 1.

***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

Read:

- The direct effects confirm **H1**: higher short-term rates reduce loan granting and higher GDP growth increases it.
- The interaction terms confirm the bank balance-sheet channel: when rates rise, weak banks cut lending more than strong banks.
- The economic magnitude is large. A 100-basis-point increase in the policy-relevant short rate reduces loan granting by low-capital banks **3.9 percent more** than high-capital banks and by low-liquidity banks **7.3 percent more** than high-liquidity banks.
- The GDP interaction is strongest and most robust for **bank capital**; the liquidity channel for GDP is weaker and sometimes not statistically significant.

5.3 Table 3: Firm-month and loan fixed-effects evidence

Read:

- This is the paper's key table. With **firm-month fixed effects**, the same firm in the same month receives different lending decisions from banks with different balance-sheet strength.

Model	(1)	(2)	(3)	(4)
Macroeconomic controls (<i>i</i>)				
$\Delta IR_t \times BANK_CAPITAL_RATIO_{it-1}$	28.566*** (6.822)	30.081*** (7.737)	19.873*** (5.681)	49.477*** (16.241)
$\Delta IR_t \times BANK_LIQUIDITY_RATIO_{it-1}$	11.548*** (2.019)	12.260*** (2.286)	5.190*** (1.926)	10.412*** (4.537)
$\Delta GDP_t \times BANK_CAPITAL_RATIO_{it-1}$	-31.426*** (5.669)	-37.078*** (5.994)	-25.668*** (3.175)	-12.594 (15.468)
$\Delta GDP_t \times BANK_LIQUIDITY_RATIO_{it-1}$	-1.602 (2.035)	-1.877 (2.166)	-2.375** (1.109)	6.947 (4.945)
Bank characteristics (<i>b</i>)				
$BANK_CAPITAL_RATIO_{it-1}$	0.308* (0.184)	0.389** (0.191)	0.175* (0.093)	-0.810 (0.503)
$BANK_LIQUIDITY_RATIO_{it-1}$	-0.075 (0.066)	-0.062 (0.070)	-0.055 (0.035)	-0.489*** (0.162)
$\ln(TOTAL_ASSETS_{it-1})$	-0.001 (0.001)	-0.003*** (0.001)	0.001 (0.001)	-0.006*** (0.002)
ROA_{it-1}	1.252*** (0.233)	1.355*** (0.261)	1.434*** (0.218)	1.371** (0.533)
$DOUBTFUL_LOANS_RATIO_{it-1}$	0.158 (0.145)	0.136 (0.164)	-0.286*** (0.088)	-0.693** (0.337)
$HERFINDAHL_BY_INDUSTRY_{it-1}$	0.016 (0.015)	0.018 (0.017)	0.002 (0.014)	0.056* (0.033)
Firm characteristics (<i>f</i>)				
$\ln(1 + NUMBER_OF_MONTHS_WITH_THE_BANK_{it-1})$	0.010*** (0.001)	0.013*** (0.001)	0.011*** (0.001)	0.026*** (0.004)
Firm-month fixed effects	Yes	No	Yes	Yes
Loan fixed effects	No	Yes	No	No
Observations	328,891	263,042	427,364	55,025
Number of bank-month clusters	8,714	8,310	10,637	6,022
Sample period	2002:02–2008:12	2002:02–2008:12	2002:02–2010:06	2002:02–2008:12

Notes: The table reports the estimated coefficients and robust standard errors (in parentheses) clustered at the bank-month level from linear probability models estimated using least squares. The sample employed in Model (4) includes only borrowers that are noncurrent to all banks. Fixed effects are included ("yes") or not included ("no"). The set of month fixed effects includes a fixed effect for every (but one) year-month during the sample period. The variable definitions and summary statistics are in Table 1.

***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

- The estimates remain close to the baseline. A 100-basis-point increase in the short rate lowers loan granting by low-capital banks about **3.8 percent more** than high-capital banks and by low-liquidity banks about **6.2 percent more** than high-liquidity banks.
- The **loan fixed-effects** specification tells the same story, showing that the result is not an artifact of borrower composition.
- The table is what supports the paper's strongest claim: under tighter conditions, weak bank balance sheets cause a true **credit crunch**.

5.4 Table 4: Summary statistics for the substitution tests

Variable	Units	Definition	Mean	SD	Min	Median	Max
<i>Dependent variable</i>							
$FUTURE_LOAN_APPLICATION_IS_GRANTED_FOLLOWING_REJECTION_{it}$	0/1	=1 if at least one future loan application by a firm is approved and the loan is granted by a bank given that a loan application was rejected during the previous three months, =0 otherwise	0.37	0.48	0	0	1
<i>Independent variables</i>							
Macroeconomic conditions (<i>i</i>)							
ΔIR_t	%	Annual change of Spanish 3-month interbank interest rates	0.26	0.82	-1.56	0.43	1.41
ΔGDP_t	%	Annual change of Spanish gross domestic product in real terms	3.01	1.09	-0.85	3.35	3.98
ΔCPI_t	%	Annual change of Spanish Consumer Price Index	3.35	0.84	1.43	3.40	5.27
Bank characteristics (<i>b</i>)							
$BANK_CAPITAL_RATIO_{it-1}$	%	The ratio of bank equity over total assets of the bank	5.31	2.04	0.00	4.83	61.75
$BANK_LIQUIDITY_RATIO_{it-1}$	%	The ratio of liquid assets (cash and balance with central banks, and loans and advances to governments and credit institutions) held by the bank over the total assets of the bank	16.50	7.86	0.04	15.52	91.62
$\ln(TOTAL_ASSETS_{it-1})$	—	The log of the total assets of the bank	17.36	1.45	10.06	17.56	20.76
$TOTAL_ASSETS_{it-1}$	000,000,000 EUR	The total assets of the bank	76.50	91.00	0.02	42.40	1,030.00
ROA_{it-1}	%	The total net income over assets of the bank	0.93	0.55	-8.93	0.90	11.92
$DOUBTFUL_LOANS_RATIO_{it-1}$	%	The doubtful loan ratio of the bank	0.90	0.92	0.00	0.61	17.36
$HERFINDAHL_BY_INDUSTRY_{it-1}$	%	The Herfindahl-Hirschman index of the bank's credit portfolio by industry	27.21	9.21	12.77	24.15	87.94
Average approached bank characteristics (<i>b</i>)							
$BANK_CAPITAL_RATIO_{it-1}$	%	The ratio of bank equity over total assets of the bank	5.25	1.89	0.00	4.77	61.75
$BANK_LIQUIDITY_RATIO_{it-1}$	%	The ratio of liquid assets (cash and balance with central banks, and loans and advances to governments and credit institutions) held by the bank over the total assets of the bank	16.37	7.48	0.04	15.53	91.37
$\ln(TOTAL_ASSETS_{it-1})$	—	The log of the total assets of the bank	17.49	1.39	10.14	17.69	20.76
$TOTAL_ASSETS_{it-1}$	000,000,000 EUR	The total assets of the bank	80.80	91.20	0.03	48.20	1040.00
ROA_{it-1}	%	The return on assets of the bank	0.93	0.50	-8.93	0.89	11.92
$DOUBTFUL_LOANS_RATIO_{it-1}$	%	The doubtful loan ratio of the bank	0.96	1.01	0.00	0.62	31.24
$HERFINDAHL_BY_INDUSTRY_{it-1}$	%	The Herfindahl-Hirschman index of the bank's credit portfolio by industry	27.10	8.69	13.88	24.45	86.67

Variable	Units	Definition	Mean	SD	Min	Median	Max
<i>Dependent variable</i>							
$FUTURE_LOAN_APPLICATION_IS_GRANTED_AFTER_EARLIER_APPLICATION_IS_MADE_{it}$	0/1	=1 if at least one future loan application by a firm is approved and the loan is granted by a bank given that a loan application was made during the previous three months, =0 otherwise	0.79	0.41	0	1	1
<i>Independent variables</i>							
Macroeconomic conditions (<i>i</i>)							
ΔIR_t	%	Annual change of Spanish 3-month interbank interest rates	0.18	0.83	-1.56	0.11	1.41
ΔGDP_t	%	Annual change of Spanish gross domestic product in real terms	3.13	0.92	-0.85	3.35	3.98
ΔCPI_t	%	Annual change of Spanish Consumer Price Index	3.32	0.77	1.43	3.40	5.27
Bank characteristics (<i>b</i>)							
$BANK_CAPITAL_RATIO_{it-1}$	%	The ratio of bank equity over total assets of the bank	5.66	1.27	0.00	5.49	50.03
$BANK_LIQUIDITY_RATIO_{it-1}$	%	The ratio of liquid assets (cash and balance with central banks, and loans and advances to governments and credit institutions) held by the bank over the total assets of the bank	16.88	5.35	0.04	16.22	91.21
$\ln(TOTAL_ASSETS_{it-1})$	—	The log of the total assets of the bank	18.07	0.88	9.91	18.27	19.90
$TOTAL_ASSETS_{it-1}$	000,000,000 EUR	The total assets of the bank	91.90	56.30	0.02	86.20	437.00
ROA_{it-1}	%	The total net income over assets of the bank	1.02	0.33	-8.93	1.00	5.25
$DOUBTFUL_LOANS_RATIO_{it-1}$	%	The doubtful loan ratio of the bank	0.81	0.60	0.00	0.63	20.50
$HERFINDAHL_BY_INDUSTRY_{it-1}$	%	The Herfindahl-Hirschman index of the bank's credit portfolio by industry	26.07	6.21	13.44	25.30	81.02
Selected firm characteristics (<i>i</i>)							
$NUMBER_OF_BANK_RELATIONSHIPS_{it-1}$	—	The number of bank relationships of the firm	3.66	3.47	0	3	102
$\ln(1 + NUMBER_OF_BANK_RELATIONSHIPS_{it-1})$	—	The log of the number of bank relationships of the firm	1.33	0.64	0	1.39	4.63

Notes: The number of observations equals 117,152 in the top panel and 746,341 in the bottom panel. EUR = 2008 euros.

Read:

- After a rejection, the probability that a firm gets a loan from another application within the next three months is only **37 percent**.
- After any application, the probability that the firm receives *some* bank loan within three months is about **79 percent**.
- These lower and upper margins set up the two substitution exercises in Tables 5 and 6.
- The table signals in advance that rejections are not trivially undone.

5.5 Table 5: Future loan granting after a rejection

TABLE 5—REGRESSION RESULTS, LOAN GRANTING AND PAST REJECTIONS
(Dependent variable: FUTURE LOAN APPLICATION IS GRANTED FOLLOWING REJECTION_{it})

Model	(1)	(2)
Macroeconomic controls (<i>t</i>)		
ΔIR_t	-7.083*** (1.407)	
$\Delta IR_t \times \text{BANK CAPITAL RATIO}_{t-1}$	23.892** (10.967)	17.980* (10.618)
$\Delta IR_t \times \text{BANK LIQUIDITY RATIO}_{t-1}$	23.888*** (4.175)	15.367*** (3.548)
ΔGDP_t	5.451*** (0.857)	
$\Delta GDP_t \times \text{BANK CAPITAL RATIO}_{t-1}$	-33.757*** (9.170)	-34.035*** (9.403)
$\Delta GDP_t \times \text{BANK LIQUIDITY RATIO}_{t-1}$	0.169 (2.944)	-0.070 (2.549)
ΔCPI_t	-1.363** (0.553)	
Bank characteristics (<i>b</i>)		
$\text{BANK CAPITAL RATIO}_{t-1}$	0.084 (0.095)	0.035 (0.093)
$\text{BANK LIQUIDITY RATIO}_{t-1}$	-0.019 (0.027)	-0.016 (0.027)
$\ln(\text{TOTAL ASSETS}_{t-1})$	0.001 (0.001)	0.001 (0.001)
ROA_{t-1}	-0.052 (0.270)	0.248 (0.278)
$\text{DOUBTFUL LOANS RATIO}_{t-1}$	-0.362 (0.227)	-0.240 (0.240)
$\text{HERFINDAHL BY INDUSTRY}_{t-1}$	0.034 (0.028)	0.040 (0.025)
Average approached bank characteristics (<i>b</i>)		
$\text{BANK CAPITAL RATIO}_{t-1}$	1.083*** (0.287)	1.051*** (0.294)
$\text{BANK LIQUIDITY RATIO}_{t-1}$	-0.016 (0.089)	-0.002 (0.077)
$\ln(\text{TOTAL ASSETS}_{t-1})$	0.037*** (0.002)	0.036*** (0.002)
ROA_{t-1}	0.952** (0.437)	1.339*** (0.417)
$\text{DOUBTFUL LOANS RATIO}_{t-1}$	1.038*** (0.341)	1.061*** (0.306)
$\text{HERFINDAHL BY INDUSTRY}_{t-1}$	0.111*** (0.029)	0.094*** (0.030)

TABLE 5—REGRESSION RESULTS, LOAN GRANTING AND PAST REJECTIONS
(Dependent variable: FUTURE LOAN APPLICATION IS GRANTED FOLLOWING REJECTION_{it}) (Continued)

Model	(1)	(2)
Selected firm characteristics (<i>i</i>)		
$\ln(1 + \text{NUMBER OF BANK RELATIONSHIPS}_{t-1})$	-0.164*** (0.010)	-0.165*** (0.010)
Other firm (<i>i</i>), industry (<i>s</i>), and province (<i>p</i>) characteristics, constant	Yes	Yes
Firm fixed effects	Yes	Yes
Month fixed effects	No	Yes
Observations	117,152	117,152
Number of month clusters	83	83
Period	2002:02–2008:12	2002:02–2008:12

Notes: The table reports the estimated coefficients and robust standard errors (in parentheses) clustered at the monthly level from linear probability models estimated using least squares. The set of included bank (*b*), firm (*i*), industry (*s*), and province (*p*) characteristics is the same as in Table 2. Fixed effects are included ("yes") or not included ("no"). The set of month fixed effects includes a fixed effect for every (but one) year:month during the sample period. The variable definitions and summary statistics of selected variables are in Table 1.

***Significant at the 1 percent level.

**Significant at the 5 percent level.

*Significant at the 10 percent level.

Read:

- Firms that are rejected initially do *not* fully neutralize that shock by applying elsewhere.
- Higher interest rates and lower GDP growth reduce the probability that a later application succeeds.
- The deterioration is stronger when the relevant banks are weakly capitalized or illiquid.
- So the bank lending channel is not just about which bank lends; it also reduces the borrower's ability to find substitutes after being turned down.

5.6 Table 6: Future loan granting after an earlier application and the role of relationships

Read:

TABLE 6—REGRESSION RESULTS, LOAN GRANTING AND PAST APPLICATIONS
(Dependent variable: FUTURE LOAN APPLICATION IS GRANTED AFTER EARLIER APPLICATION IS MADE_{it})

Model	(1)	(2)	(3)	(4)
Macroeconomic controls (<i>i</i>)				
ΔIR_t	-3.184*** (0.743)		-5.781*** (1.344)	
$\Delta IR_t \times \text{BANK CAPITAL RATIO}_{it-1}$	12.337* (7.007)	12.486* (6.942)	39.628** (16.046)	37.993** (15.600)
$\Delta IR_t \times \text{BANK LIQUIDITY RATIO}_{it-1}$	7.491*** (2.146)	6.643*** (1.679)	14.740*** (4.310)	13.292*** (3.995)
ΔGDP_t	4.968*** (0.545)		8.162*** (1.156)	
$\Delta GDP_t \times \text{BANK CAPITAL RATIO}_{it-1}$	-35.679*** (5.784)	-31.473*** (5.431)	-87.526*** (17.132)	-83.131*** (16.768)
$\Delta GDP_t \times \text{BANK LIQUIDITY RATIO}_{it-1}$	-7.539*** (2.283)	-3.200** (1.486)	0.757 (4.639)	4.614 (4.260)
ΔCPI_t	-0.078 (0.123)		-0.099 (0.120)	
Bank characteristics (<i>b</i>)				
$\text{BANK CAPITAL RATIO}_{it-1}$	1.284*** (0.198)	1.061*** (0.188)	2.988*** (0.592)	2.844*** (0.577)
$\text{BANK LIQUIDITY RATIO}_{it-1}$	0.224*** (0.080)	0.041 (0.054)	-0.109 (0.153)	-0.229 (0.139)
$\ln(\text{TOTAL ASSETS}_{it-1})$	0.003*** (0.001)	0.007*** (0.001)	0.003*** (0.001)	0.007*** (0.001)
ROA_{it-1}	0.171 (0.203)	0.691*** (0.196)	0.111 (0.203)	0.624*** (0.198)
$\text{DOUBTFUL LOANS RATIO}_{it-1}$	0.282 (0.182)	0.132 (0.179)	0.222 (0.183)	0.098 (0.181)
$\text{HERFINDAHL BY INDUSTRY}_{it-1}$	-0.026 (0.020)	0.073*** (0.020)	-0.013 (0.019)	0.075*** (0.020)
Selected firm characteristics (<i>i</i>)				
$\ln(1 + \text{NUMBER OF BANK RELATIONSHIPS}_{it-1})$	-0.002 (0.003)	-0.001 (0.003)	0.059** (0.023)	0.070*** (0.024)
Interactions with $\ln(1 + \text{NUMBER OF BANK RELATIONSHIPS}_{it-1})$				
$\Delta IR_t \times \ln(1 + \text{NUMBER OF BANK RELATIONSHIPS}_{it-1})$			2.193*** (0.681)	2.004** (0.827)
$\Delta IR_t \times \text{BANK CAPITAL RATIO}_{it-1} \times \ln(1 + \text{NUMBER OF BANK RELATIONSHIPS}_{it-1})$			-21.042** (9.284)	-19.165** (9.320)
$\Delta IR_t \times \text{BANK LIQUIDITY RATIO}_{it-1} \times \ln(1 + \text{NUMBER OF BANK RELATIONSHIPS}_{it-1})$			-6.127** (2.517)	-5.780* (2.959)
$\Delta GDP_t \times \ln(1 + \text{NUMBER OF BANK RELATIONSHIPS}_{it-1})$			-2.304*** (0.657)	-2.275*** (0.711)
$\Delta GDP_t \times \text{BANK CAPITAL RATIO}_{it-1} \times \ln(1 + \text{NUMBER OF BANK RELATIONSHIPS}_{it-1})$			39.880*** (11.314)	39.325*** (11.185)
$\Delta GDP_t \times \text{BANK LIQUIDITY RATIO}_{it-1} \times \ln(1 + \text{NUMBER OF BANK RELATIONSHIPS}_{it-1})$			-6.113** (2.805)	-5.908** (2.870)
$\ln(1 + \text{NUMBER OF BANK RELATIONSHIPS}_{it-1}) \times \text{BANK CAPITAL RATIO}_{it-1}$			-1.295*** (0.391)	-1.355*** (0.384)
$\ln(1 + \text{NUMBER OF BANK RELATIONSHIPS}_{it-1}) \times \text{BANK LIQUIDITY RATIO}_{it-1}$			0.263*** (0.089)	0.215** (0.089)

TABLE 6—REGRESSION RESULTS, LOAN GRANTING AND PAST APPLICATIONS
(Dependent variable: FUTURE LOAN APPLICATION IS GRANTED AFTER EARLIER APPLICATION IS MADE_{it})
(Continued)

Model	(1)	(2)	(3)	(4)
Other firm (<i>i</i>), industry (<i>s</i>), and province (<i>p</i>) characteristics, constant	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes
Month fixed effects	No	Yes	No	Yes
Observations	746,341	746,341	746,341	746,341
Number of month clusters	83	83	83	83
Period	2002:02–2008:12	2002:02–2008:12	2002:02–2008:12	2002:02–2008:12

Notes: The table reports the estimated coefficients and robust standard errors (in parentheses) clustered at the monthly level from linear probability models estimated using least squares. The set of included bank (*b*), firm (*i*), industry (*s*), and province (*p*) characteristics is the same as in Table 2. Fixed effects are included ("yes") or not included ("no"). The set of month fixed effects includes a fixed effect for every (but one) year:month during the sample period. The variable definitions and summary statistics of selected variables are in Table 1.

*** Significant at the 1 percent level.

** Significant at the 5 percent level.

* Significant at the 10 percent level.

- Even when the dependent variable is broadened to whether the firm gets *any* bank loan after applying, tighter conditions still reduce credit availability.
- The interaction between macro conditions, bank weakness, and the number of bank relationships is economically important.
- Firms with multiple bank relationships are better insulated. For firms with about **three bank relationships**, the difference between weak and strong banks is about **one-third** of the difference for firms with no current bank relationship.
- This shows that relationship banking mitigates but does not eliminate the bank balance-sheet channel.

5.7 Robustness and extensions

Read:

- The paper is mainly **table-driven**; there are no main figures carrying the empirical argument.
- The results survive alternative fixed-effects structures, logit models, different matching windows, and the sample extension through 2010:06.
- They also survive alternative ways of handling bank capital and broader interaction sets.
- Unreported intensive-margin evidence using all granted loans points in the same direction, reinforcing that the mechanism is not unique to the application margin.

6 Short summary

- **Conclusion.** Tighter monetary policy and weaker economic conditions reduce loan supply, and they do so disproportionately through banks with weaker balance sheets.
- **Identification insight.** Loan applications plus firm-month or loan fixed effects allow the paper to separate loan supply from demand more cleanly than earlier work.
- **Empirical lesson.** Weak bank capital and liquidity matter for the transmission of macro shocks to credit, and borrowers cannot costlessly insulate themselves by switching banks.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper establishes a strong and credible bank lending channel of monetary policy using loan-application data.
- It shows that bank balance-sheet strength is not a passive correlational feature but an active transmission mechanism through which macro shocks affect credit supply.
- It also shows that the resulting supply restrictions are binding for firms, because they cannot be fully offset through substitution to other lenders.

7.2 Economic intuition

- When short-term rates rise or growth falls, the cost of external finance for banks increases and the value of internal liquidity rises.
- Banks with less capital or liquidity face a more acute tightening in effective funding conditions and therefore become more reluctant to approve new loans.
- Borrowers feel this most strongly when they lack alternative relationships, so the contraction in supply becomes a real contraction in firm access to credit.

7.3 Takeaway

This paper is best understood as a clean micro-foundation for the bank balance-sheet channel. Its central message is that to understand how monetary policy and business-cycle conditions affect credit, it is not enough to observe loans or banks in the aggregate. One must observe *applications*, compare how different banks respond to the *same borrower at the same time*, and then ask whether the borrower can substitute away from weak lenders. Once the paper does this, the evidence is clear: weak bank balance sheets amplify macro tightening into a genuine credit crunch.